

DSA Day 6: Single Linked List - build & traverse

Book: Data Structures Through C in Depth - Ch3 §3.1.1-3.1.4 · pp 48-58 | Code in Python

Overview

A linked list chains nodes, each holding data and a reference to the next. No random access, but $O(1)$ insert/delete once you have the position.

Key concepts to master

- Node(data,next), head, None end
- Traversal is $O(n)$
- Searching a list
- Building by appending

Python implementation

```
class Node:
    def __init__(self,data): self.data=data; self.next=None
def build(vals):
    head=None
    for v in reversed(vals):
        n=Node(v); n.next=head; head=n
    return head
def find(head,key):
    while head:
        if head.data==key: return True
        head=head.next
    return False
```

Complexity

Access/search $O(n)$; insert/delete at known node $O(1)$.

Common pitfalls

- Losing the head reference
- Dereferencing None

Practice

- Build 1->2->3->4 and print
- Search for a value

